

- A flexible query language to leverage this dimensionality
- Non-reliance on distributed storage; single server nodes are autonomous
- Time series collection happens via a pull model over HTTP
- Pushing time series is supported via an intermediary gateway
- Targets are discovered via service discovery or static configuration
- Multiple modes of graphing and dashboarding support

## Docker Compose

Another useful tool for developers is Docker Compose. The prime importance of this tool is that it provides a base for running multi-container Docker applications, along with its dependencies, in single file. Also, applications can run with a single command. All the applications are defined in a YAML file, which comprises all the options used in 'Docker run'.

Docker Compose is a three-step process:

- Defines your app's environment with a Docker file so it can be reproduced anywhere.
- Defines the services that make up your app in *docker-compose.yml*, so that they can be run together in an isolated environment.
- Lastly, runs *docker-compose up* and Compose will start and run your entire app.

## Kitematic

Kitematic is an open source project designed to simplify and streamline using Docker on a Mac or Windows system. Kitematic aims to make Docker useful as a desktop-environment developer's tool. It makes the process of downloading Docker images, spinning them up, and managing them into a task no more difficult than, say, using VMs in an application like VMware Workstation.

The Kitematic GUI launches from the home screen and presents curated images that can run instantly. The user can search for any public image on Docker Hub from Kitematic just by typing in the search bar. The GUI is used to create, run and manage containers just by clicking on buttons. Kitematic allows you to switch back and forth between the Docker CLI and the GUI. Kitematic also automates advanced features such as managing ports and configuring volumes. Users can use Kitematic to change environment variables, stream logs, and single-click terminals into your Docker container -- all from the GUI.

The features of Kitematic include:

- **Fast and easy set-up:** One-click install gets Docker running on your Mac, and lets users control app containers from the GUI.
- **Docker Hub integration:** Allows users to easily search favourite images on Docker Hub from Kitematic GUI to create and run your app containers.

- **Seamless experience between the CLI and GUI:** Enables users to seamlessly switch between Kitematic GUI or Docker CLI to run and manage your application containers.
- **Advanced features:** Automatically maps ports and visually changes environment variables, configuring volumes, streamline logs and CLI access to containers.

## Logspout

Logspout is a great tool for managing logs generated by programs running inside Docker containers. It manages the log router for the Docker container by attaching all containers to a host, and consists of an extensible module system.

The size of Logspout is just 14MB and it has Busybox at its core, which enables developers to route all the logs of Docker containers to a central location like a single JSON object.

## Flocker

Flocker is an open source container data volume orchestrator for Dockerised applications. It migrates data along with containers as they change hosts. Flocker gives users the tools they need to run containerised services like databases in production. Flocker manages Docker containers and data volumes together.

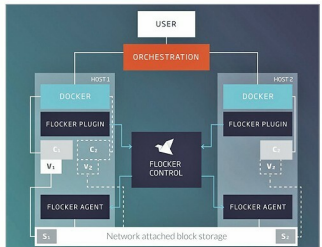


Figure 1: Flocker

Flocker is designed to work with the other tools to build and run distributed applications. It can be used with popular container managers or orchestration tools like Docker, Kubernetes, Mesos, etc. Flocker supports block-based shared storage such as Amazon EBS or OpenStack Cinder to choose the storage back-end that is best for an application.

Its features are:

- Seamless database migrations for micro services
- Integrates into existing Docker workflow